Ayaan Sulman

AI / ML Engineer

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Summary

Software engineer focused on applied AI: LLM agents with tool calling, local and on-device models, and ML pipelines for video, audio and images, taken from benchmark to shipped product in Python, PyTorch and TypeScript.

Experience

ElkTech — Associate Software Engineer Jan 2026 – Present

- **Kira AI**: AI assistant for web, desktop, iOS and Android (5-person team), showcased at the Dubai AI Festival.
  - **Offline AI:** benchmarked local LLMs and speech models for the desktop app and shipped Qwen 3.5 4B (llama.cpp, 4-bit) with Whisper large-v3-turbo: 48 of 48 agent tasks in 4 languages at ∼1.4 s each with no invented actions, and 2–10% word error rate at ∼1 s per voice command.
  - **AI agents:** built project boards where a flow engine hands tickets to agent workers (research, writing, coding, browsing, calling), plus chat and voice tools that start goals and sales deals.
  - **Agent reliability:** made chat and voice claim an action only if a tool really ran, added approvals before changes, and gave every agent action a receipt in a Merkle log stamped on Base (Ethereum L2).
  - **Assistant features:** Meeting Autopilot (meeting follow-ups sent with one approval), one search across meetings, calendar, mail, files and memory, and short tool-using tasks run on a user's behalf.
- **Naito**: Instagram Reels analytics for a French client; main developer of its ML pipeline.
  - Built video analysis with whisper.cpp transcripts, librosa audio features, and thumbnail attention heatmaps (MSI-Net saliency) and quality scores (NIMA), run with PyTorch and ONNX Runtime.
  - Built topic modeling over captions and transcripts (multilingual Model2Vec embeddings, KMeans, c-TF-IDF labels, PCA map) and a virality score that adjusts to the evidence available.
- **Lean AI Factory**: toolkit for on-device AI in iOS apps. Curated 30 open-source vision, audio, NLP and forecasting models; built Skill Studio, which tunes coding-agent skills with local LLMs (DeepSeek-R1, Qwen2.5-Coder via Ollama), and a saliency-based tool that predicts where users look on app screens.
- **Helpath**: platform for UK domestic-abuse refuges; built frontend and backend features in Next.js and NestJS.

US Mobile — Product Support Engineer Oct 2025 – Dec 2025

- Resolved network and account issues for 4+ customers at once with a 98.2% CSAT; ranked 2nd of 20 in training.

Big Entities — Software Engineering Intern Jul 2025 – Oct 2025

- Backend engineer (.NET Core) on **FieldForce AI**, an analytics platform used by 2,000+ staff at Pharmasol and Biocare, and **Healers**, a Canadian telehealth app; managed SQL Server and PostgreSQL databases.

Clicksotic — Front End Developer Mar 2024 – Feb 2025

- Built React frontends for the Caroline School of Law learning management system and **DripSaint**, a custom-apparel store with an AI image generator for designing prints.

Exputer — Gaming Content Writer (Part-time) Jul 2023 – Aug 2024

- Published 30+ game guides on Fortnite, LEGO Fortnite and Minecraft, and moderated the site's forums.

Projects

- **Mood Flicks** (*Llama 3, NLP*): recommends films by using Llama 3 to read a user's mood from free text.
- **ObscuraNet** (Final Year Project, team of 4; *Python*): serverless P2P messaging with onion routing, ChaCha20-Poly1305, proof-of-work Sybil resistance and dummy traffic; tested on 100 Docker nodes, under 2 s delivery.

Technical Skills

- **ML & Vision/Audio:** PyTorch, Hugging Face Transformers, ONNX Runtime, OpenCLIP, scikit-learn, OpenCV, MediaPipe, librosa, NumPy, TensorFlow
- **LLMs & Agents:** tool calling, agent workflows, model evaluation and benchmarking, llama.cpp, Ollama, Whisper / whisper.cpp, Qwen, Llama 3, embeddings (Model2Vec)
- **Languages:** Python, TypeScript, JavaScript, C#, SQL, Solidity
- **Apps & APIs:** FastAPI, Django, NestJS, Node.js, .NET, React, Next.js, React Native, Electron
- **Data & DevOps:** PostgreSQL, SQL Server, MongoDB, Prisma, Redis, Docker, Nginx, Git
- **Tools:** Claude Code, Codex, Playwright **Spoken:** English, Urdu (both fluent)

Education

Forman Christian College (A Chartered University), Lahore — BS Computer Science 2021 – 2025

Awards & Leadership

- 1st of 80, FCCU GDSC Programming Competition (2023); Best Project nominee, FCCU COMSPEX (2025)
- Co-Lead, Registration Team, Ignite FCCU (2023); Lead, Registration Team, Omnilife VR/AR Society (2022)